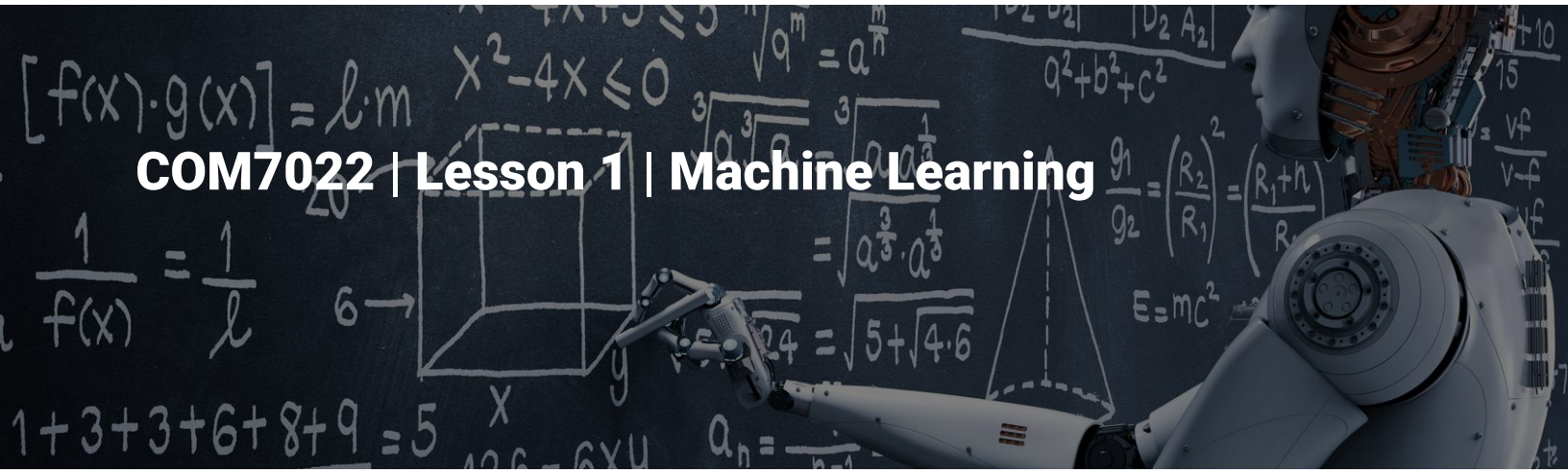


COM7022 | Lesson 1 | Machine Learning

Machine Learning (COM7022)

Lesson 1: Introduction to Machine Learning

Cover image by Phonlamai Photo from Shutterstock.com

- **Lesson introduction**
- **Before you get started**
- **Topic 1: What is machine learning?**
- **Topic 2: Toolbox**
- **Topic 3: NumPy and Pandas**
- **Topic 4: Scikit-Learn and SciPy**
- **Knowledge check**

Lesson summary

Reading List

Lesson introduction

Welcome to Lesson 1

LESSON INTRODUCTION | LEARNING OUTCOMES | ACCESSIBILITY GUIDANCE

Welcome to Lesson 1 on Machine Learning!

Lesson 1 is designed to provide a comprehension introduction to the exciting world of machine learning. In this module, we are particularly interested in the practicality and implementations of machine-learning algorithms and techniques in data science.

In the era of data explosion, one of the challenges of a data user is to obtain the necessary insights into the data collected and to use the knowledge learned for the benefit of the company or organisation.

Lesson 1 has been designed to give you a concise introduction to the world of machine-learning practitioners. This lesson will provide you with the necessary fundamental understanding of machine learning and the tools you will be using throughout this module. It is a challenging module, but balancing the challenge with enjoyment will make it a rewarding experience.

The following are the topics that we will cover in this lesson:

- **Topic 1 – What is machine learning?**
  This topic will provide you with a brief introduction to machine learning, and to the definitions and, most importantly, real-world applications of machine learning.
- **Topic 2 – Why Python?**
  This topic will examine the rationale behind selecting Python as the programming language of choice in this module.
- **Topic 3 – NumPy and Pandas**
  We will review NumPy and Pandas as part of library packages that will be used when we implement machine learning models.
- **Topic 4 – Scikit-learn and Scipy**
  Finally, we will be looking at Scikit-learn and Scipy, two of the most widely used machine-learning library packages in Python for a starter practitioner of the 'art'.

All topics included within this lesson will hopefully provide you with the motivations and an initial glimpse into the world of a machine-learning practitioner.



1. Demonstrate a practical understanding of the characteristics of various machine-learning methods and their applications in data analysis.
2. Select and apply appropriate machine-learning techniques for a given scenario.
3. Critically evaluate the results from the application of machine-learning techniques and propose improvements.

Lesson 1 aligns with Learning Objective 1: **Demonstrate a practical understanding of the characteristics of various machine-learning methods and their applications in data analysis.**



Computing

At Arden University

We have worked across teams at Arden University to try and create an accessible and inclusive experience within the learning content. We are continuously assessing the accessibility of our lessons to make improvements and enhancements based on testing and feedback.

This lesson may include screenshots of code and software used. Please be aware that these may not always have a text alternative, and that other features such as size and colour may not be accessible.

If you experience any issues with accessing the learning content throughout this module, there are several support services in place to help you.

- Arden University has a dedicated Inclusion Services Team who can help you with getting adjustments and equipment.
- The Library Portal also has information on the accessibility tools that can help you access resources: Accessibility Tools – Library Portal
- If you are struggling with accessing a specific part of the content e.g. a video resource that is missing captions, please speak to your Lecturer who may be able to advise on an alternative resource in the interim to ensure you are able to get the learning you need to progress with the lesson.

Continue to Before you get started

Before you get started



Image from Articulate Rise 360 Content Library.

How much of the data we create do you think is analysed and used?

Move the slider to match your estimate, then select submit.



0 %

Submit

Reveal

Only 0.5% of all data we create is ever analysed and used (Burn-Murdoch, 2012).

Continue

Did you know?

Who do you think is credited with coining the term ‘machine learning’?

Continue

Arthur Samuel from IBM is credited with coining the term ‘machine learning’ in his research on the game of checkers (Samuel, 1959).



Arthur Samuel playing checkers against the IBM 701, February 24, 1956. Image by IBM.

Continue

Which business sector do you think is the most profitable to implement machine learning?

Continue



Image from Articulate Rise 360 Content Library.

The most profitable area to implement machine learning is in sales and marketing (Glackin and Adivar, 2023).

Continue to Topic 1

Topic 1: What is machine learning?



Image from Articulate Rise 360 Content Library.

In a world where data is collected daily, from how you like a post on Facebook or Instagram to your shopping habits in your favourite supermarket, it is clear that we live in the data age where data explosions have occurred at an exponential rate.

With millions of data collected daily, we need versatile and powerful tools that will allow us to discover the hidden 'gems' within the data. The processes need to be automatic and able to deal

with large numbers of data instances in a short period of time. This has given birth to data mining, and in this module we will focus on implementing machine-learning algorithms in data mining.

So, what is machine learning?

Machine learning (ML) is a branch of artificial intelligence that focuses on simulating human learning behaviours towards the data presented and improving its performance and accuracy accordingly. It can make predictions based on common patterns hidden within datasets, identifying similar patterns within a dataset, and organising them into groups or clusters.

In general, machine learning can be split into three main categories:

1. SUPERVISED LEARNING | **2. UNSUPERVISED LEARNING** | **3. REINFORCEMENT LEARNING**

This describes a class of problems where an ML model is developed to learn the mapping of data instances to a known class/target. In this module, we will link this type of learning to classification. An example is a model for approving house mortgage applications, in which the model 'learns' to approve or disapprove (target), based on a historical dataset collected.

1. SUPERVISED LEARNING | **2. UNSUPERVISED LEARNING** | **3. REINFORCEMENT LEARNING**

This describes problems where an ML model is developed to explore the patterns within the datasets, allowing the model to form groups or clusters based on similarities and dissimilarities between the data instances. In this case, the model is not provided with a known class for the model to learn. As such, unsupervised learning does not have a 'teacher' correcting the model, as in the case of supervised learning. In this module, we will

link this type of learning to clustering. An example is a new marketing campaign where surveys are collected and a model is needed to group/cluster the potential customers, without having any prior data to refer to.

1. SUPERVISED LEARNING	2. UNSUPERVISED LEARNING	3. REINFORCEMENT LEARNING

Unlike supervised or unsupervised learning, reinforcement is the development of an ML model via trial and error, where learning will halt when an optimum model is achieved and maximum reward (performance) can be achieved. This can be easily understood with the analogy of teaching a puppy how to sit. Without verbal understanding, a reward in the form of treats will be given when a puppy takes the right action, withholding treats when it makes a wrong action.

Let us look into some examples of real-world applications of ML in data analytics:

1. **Target marketing:** ML is used to predict potential future customers to enable target marketing. Companies will use previous customer purchase behaviour and build a model to predict the possibilities of potential customers purchasing certain products.
2. **Agriculture and environmental protection:** ML has been used to assist farmers in identifying and controlling weeds to reduce their harmful impact on the environment, thereby optimising the use of herbicides. See: Real-time Weed Identification Using Machine Learning and Image Processing in Oil Palm Plantations by Firmansyah *et al.*
3. **Healthcare:**
    - Machine learning is used to predict potential cardiovascular disease using patients' histories. See A Novel Study on Machine Learning Algorithm-Based Cardiovascular Disease Prediction by Khan *et al.*

- Identifying whether a tumour is benign using medical scan images: A dataset of known benign or non-benign medical images is used to build a supervised ML model. When a new unknown medical image is used as test input to the model, the model will be able to determine the output of the diagnostic based on 'learned' experience.

4. **Financial sector:** Credit card user behaviours are used as data sources and techniques to identify whether a transaction can be deemed suspicious and whether actions should be taken.

As we can see, the type of ML we use and select is mainly based on the business objectives and needs. As ML practitioners, we need to be able to identify the questions and what ML algorithms or techniques will be our best solutions to find the answer. In addition to finding the possible answer, we must focus on the performance and accuracy of the model we develop.

Continue

Topic 1 - Activity 1

Discussion on iLearn - Data explosion



The question is:

> *How can we benefit from all the collected data and use the hidden knowledge?*

Companies and organisations from across the globe collect and generate huge datasets, from sales transactions to customer feedback. For example, we collect club card points from supermarkets such as Tesco.

Have you ever asked yourself why Tesco provides you with all the benefits of club points and what they can gain from collecting your purchase history?

Post your comments to the module's discussion forum.

Continue

Topic 1 - Activity 2

Discussion on iLearn - Applications of machine learning

Can you identify some applications that use ML in your daily life? Post your suggestions to the group's discussion forum.

Choose one of the ML applications presented above and elaborate on how the advancement of ML helps that industry.



Continue

Topic 1 - Activity 3

Reading



- Machine Learning: Algorithms, Real-World Applications and Research Directions by Sarker
- Internet of Things (IoT) Security Intelligence: A Comprehensive Overview, Machine Learning Solutions and Research Directions by Sarker *et al*.
- The Bank of England and Financial Conduct Authority surveys into the state of machine learning in UK financial services: Machine learning in UK financial services by Blake et al.
- Market Basket Analysis Using FP Growth and Apriori Algorithm: A Case Study of Mumbai Retail Store by Venkatachari and Chandrasekaran

Continue

TOPIC 1: SUMMARY

Within the first introductory topic, we have looked at what ML is, as well as at the motivations and the expansion of the area. We have also examined the types of ML categories and the fundamental differences between them. Finally, we looked at some examples of ML applications.

Now that we have a basic understanding of ML, it is time to take our first steps into the methodologies and ideologies that will propel us into becoming machine learning practitioners.

Continue to Topic 2

Topic 2: Toolbox



Image by AnshiNoWara from Sketchfab.com.

Did you know?

When do you think was Python created?

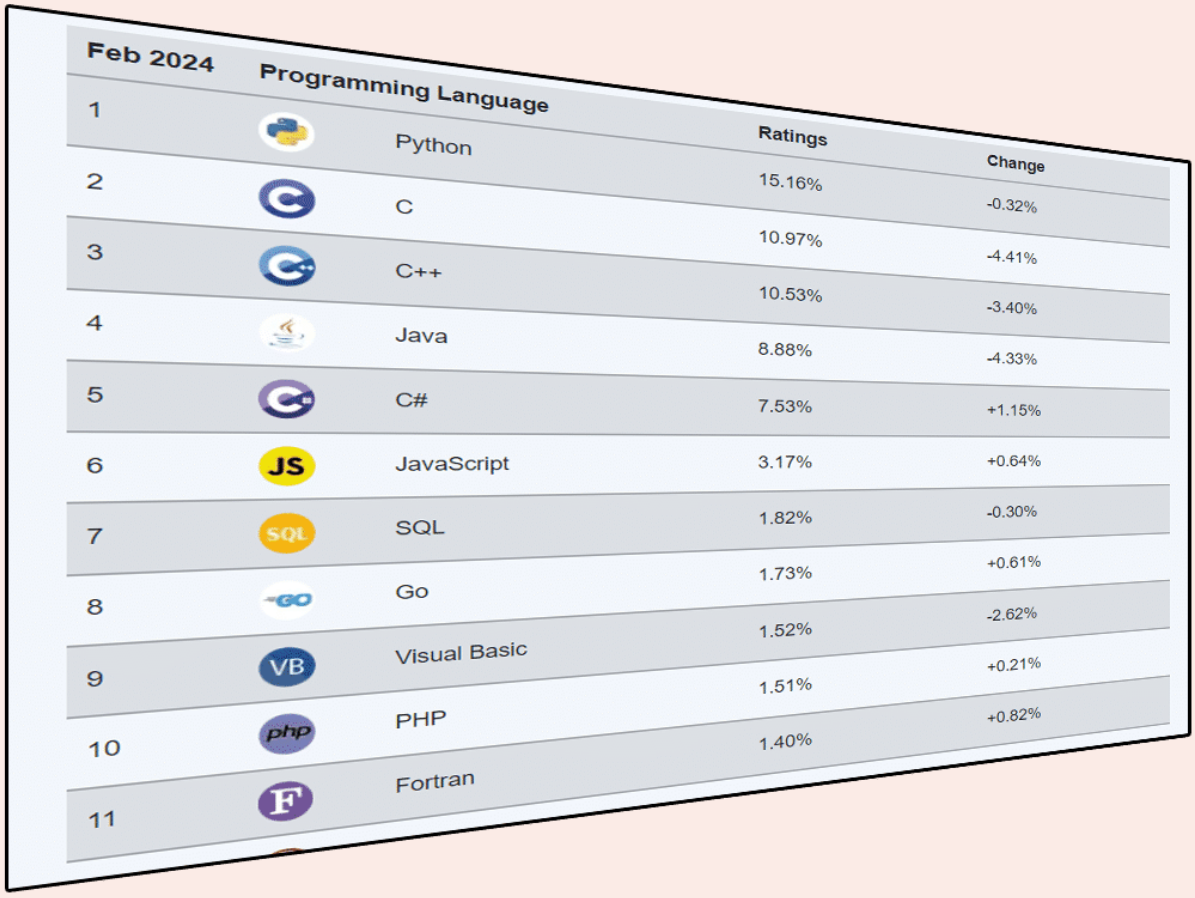
Continue

In December 1989, Python's creator Guido van Rossum was looking for a hobby to keep him occupied in the week around Christmas. He had been thinking of writing a new scripting language that would be a descendent of ABC and also appeal to Unix/C hackers. He chose to call it Python (van Rossum, no date).



Photo by Kushal Das from Flickr.com.

Continue



Feb 2024	Programming Language	Ratings	Change
1	Python	15.16%	-0.32%
2	C	10.97%	-4.41%
3	C++	10.53%	-3.40%
4	Java	8.88%	-4.33%
5	C#	7.53%	+1.15%
6	JavaScript	3.17%	+0.64%
7	SQL	1.82%	-0.30%
8	Go	1.73%	+0.61%
9	Visual Basic	1.52%	-2.62%
10	PHP	1.51%	+0.21%
11	Fortran	1.40%	+0.82%

Image by Arden University. Data from TIOBE.com.

Python has been in the top three programming languages for data science for the past few years due to the simplicity and community support available (Canales Luna, 2023).

Continue

An advantageous way to learn a new process, technique or topic is to visualise a toolbox for the essential tools required within that specific subject area – very like the way in which a builder may build a house (e.g., hammer, saw, chisel, etc.). Without such tools, the new process would be very

difficult not only to learn, but also to complete. If we were to circle back around to mapping a toolbox for IT purposes, such as building a website as a web developer, we would need to include a selection of programming languages, such as HTML, CSS and probably JavaScript. We would also require a text editor and a website builder such as WordPress, and probably a webhosting application and domain names. These are all tools, techniques and processes that need to be fully understood in order to complete the finished website. ML is no different, and for the remainder of this topic we will be demonstrating some of the essential tools that are recommended to be included in your toolbox.

Continue



Image from Articulate Rise 360 Content Library.

Data, from base principles up, is the input that we need in order not only to train an ML model, but also to make those all-important predictions. As we discussed in Topic 1, data comes in two basic

forms: structured and unstructured. As structured data is already conveniently organised and labelled, as shown in Table 1.01, it is advantageous for a relatively new ML analyst to manipulate and analyse structured data. Unstructured data, such as audio and video files, cannot be reorganised into structured rows and columns, so the analysis of it is slightly more complex.

Date	**Bitcoin Price**	**Days Transpired**
12/05/2020	184.34	2
26/11/2021	246.78	143
15/06/2022	774.01	255
04/08/2023	519.88	476
17/01/2024	1103.76	834

Table 1.01. Structured dataset by Arden University

Although you are all probably familiar with the anatomy of a structured dataset, it is advantageous to recap the anatomy again in connection to an ML environment. As Table 1 shows, tabular/structured data contains columns, and within each column is a feature, which is also commonly referred to as a variable, dimension or attribute. Whatever name we give the feature, it means the exact same thing. Conversely, each row in the table represents a single observation of a given feature. In some studies, rows are referred to as a case or value – again, whatever you wish to call them, they mean the exact same thing. To add even more complexity to naming conventions, in an ML environment columns are also referred to as vectors. Hopefully, Figure 1.01 below will visually aid in the naming convention complexity.

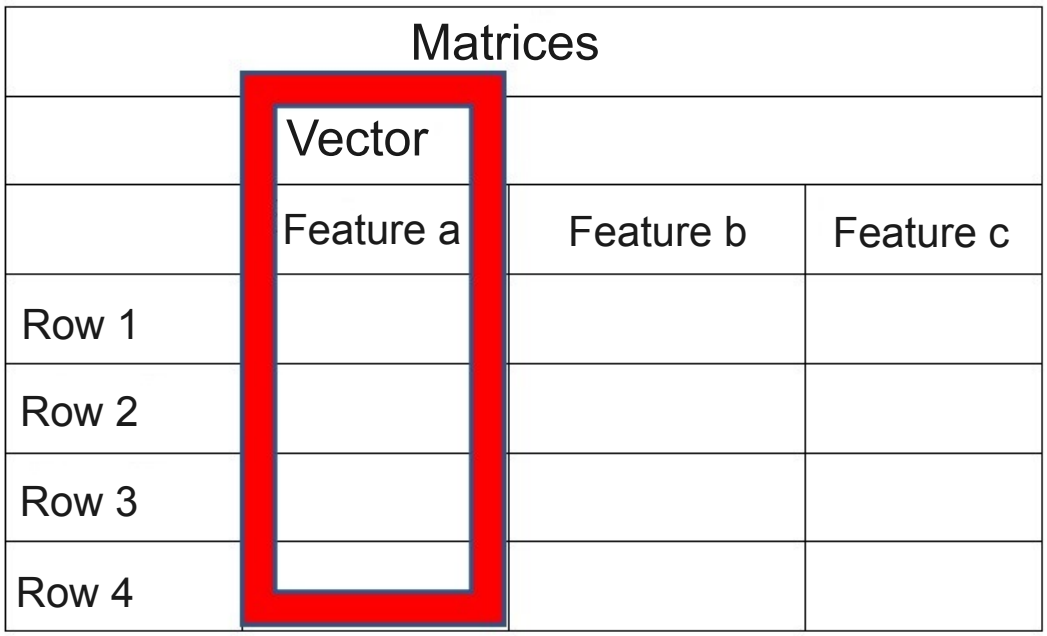


Figure 1.01. Figure by Arden University. (Accessible text).

As mentioned above, each column is known as a vector, and the vectors represent X and Y values. When there are multiple vectors, they are collectively known as matrices – we know about these operations from a previous module within this programme. Within an ML environment, Y, which is known as the dependent variable, will already exist within the dataset and is used to aid in the discovery of patterns in relation to the independent variable.

Continue

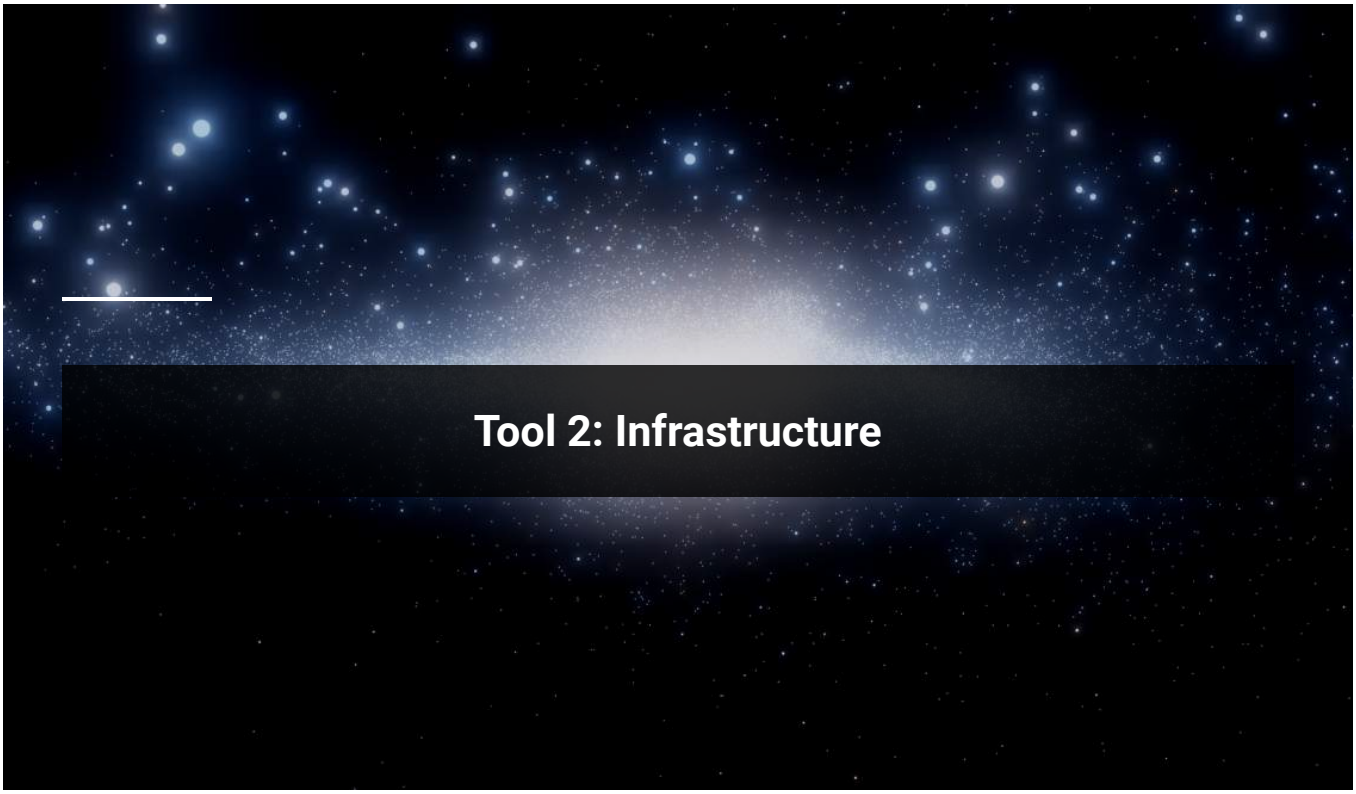


Image by Loic Norgeot from Sketchfab.com.

The second ML tool that is important to fully understand is the infrastructure of the environment. In all cases, this would include all the platforms and tools that one might need to analyse and process data. When beginning to explore ML, it will be assumed that you will be adopting and using various software platforms, and this module and the programme in its entirety is no exception. Here we will be primarily using Jupyter notebooks and Google Collaboratory as our Integrated Development Environment (IDE) so that we can write in Python. As we know, Python includes many pre-compiled libraries that aid the analyst in the pre-processing and analysis of the data, such as Numpy, Pandas and Scikit-Learn. These libraries help the user to execute complex algorithms and models with minimal use of code. It is worth spending some time becoming familiar with the multitude of pre-compiled libraries that Python has to offer novice and experienced ML analysts alike.

In addition to the pre-compiled libraries, it is highly probable that you will require more specialised libraries within Python for data visualisation purposes, such as Matplotlib and Seaborn, both of which support the creation of charts, maps and many other graphical tools. We will be fully utilising all the aforementioned libraries as we progress through this programme.

Continue

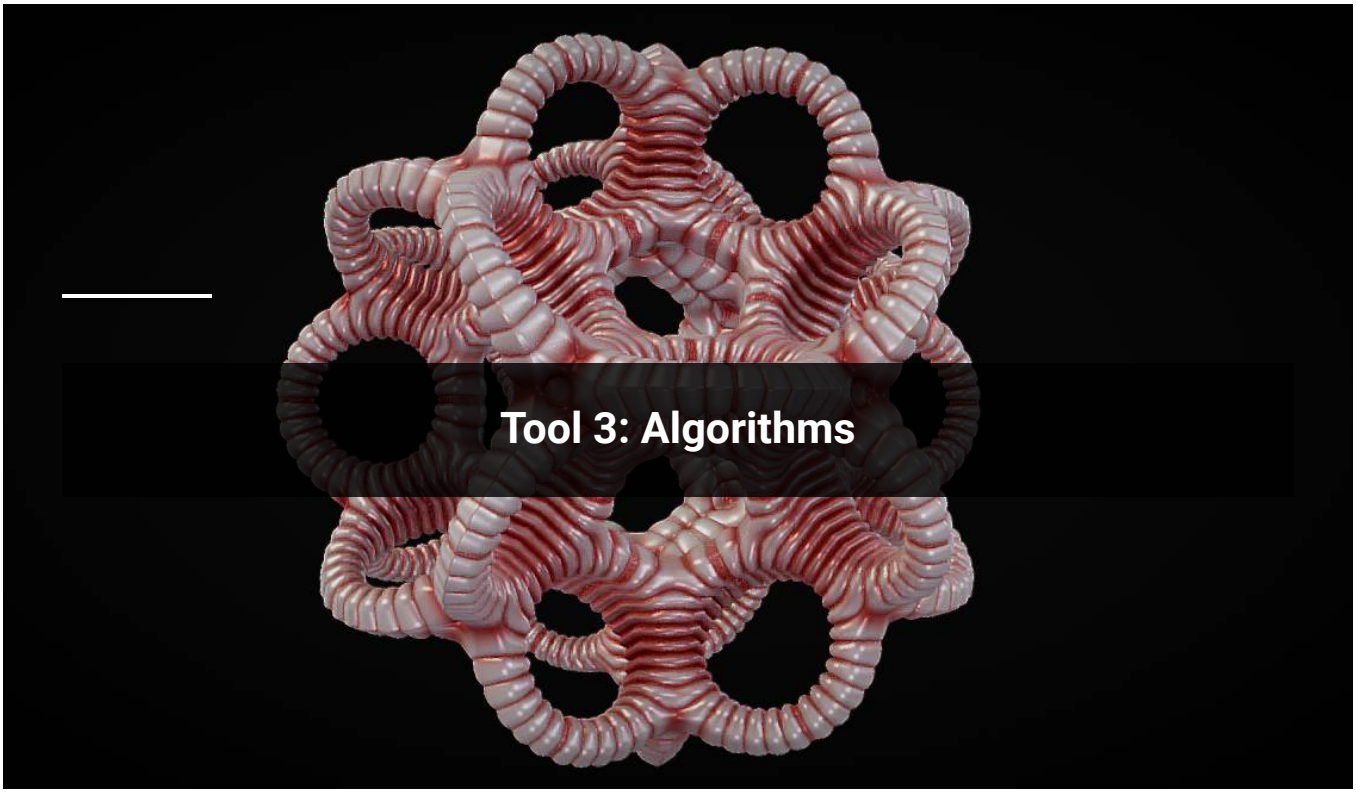


Image by Aiekick froms Sketchfab.com.

Some algorithms are pre-compiled within various Python libraries. New ML analysts tend to begin their analysis journey using simple supervised learning algorithms, as we mentioned at the beginning of this topic. Initial algorithms include regression and various types of classifications, all of which will be demonstrated within this module. When the basics in applying ML algorithms have been accomplished, the next development will be to learn unsupervised ML models, such as K-

means clustering – however, this is purely down to the individual. Whatever path you choose within your ML development, solid foundations are always a very good place to start.

It should be noted, however, that no matter how insightful your analysis may be, without the appropriate way of communicating to decision-makers it become almost irrelevant. This is why data visualisation is of the utmost importance for both new and experienced ML analysts. The ability to convey meaningful data discoveries through graphical representations cannot be underestimated as a way to achieve clear and efficient data story telling.

As you are hopefully aware, confidence in applying and understanding all the tools mentioned above is of paramount importance when beginning to understand the world of ML within a data science environment. Although Tool 1 may seem nonsensical, fully understanding the data and how this is represented numerically within a coding environment is vital.

Continue

Topic 2 - Activity 1

Discussion on iLearn - Initial Data Science Research



Research the ethical implications that are always present when engaging with an ML environment.

Write your extended answer within the group discussion forum for further debate among peers and your tutors.

Continue

The following blog by the IBM Cloud Team provides a good analysis of the differences between Python and R.

Python vs. R: What's the Difference? by IBM

What is your opinion about the programming languages discussed in the article?



Continue

Topic 2 - Activity 2

Reading



1. Pages 3-10 in Chapter 1: An Introduction to Data Mining and Predictive Analytics from Data Mining and Predictive Analytics by Larose and Larose.
2. Pages 10-35 in Chapter 1: Changing Business Environments and Decision Support Systems from Business Intelligence and Big Data: Drivers of Organizational Success by Olszak.

Continue

TOPIC 2: SUMMARY

Topic 2 has been an essential theoretical journey for the ML practitioner. Foundational material that has been discussed within this topic cannot be overestimated. Although certain items contained within this topic may be second nature at this stage in your learning, it is important to recap the essential tools that an ML analyst must become fully aware of and, more importantly, fully understand in order to build a successful career.

As mentioned at the beginning of this topic, we will look at the most popular libraries and packages throughout this module and in your journey to becoming an ML specialist.

Continue to Topic 3

Topic 3: NumPy and Pandas



Image from Articulate Rise 360 Content Library.

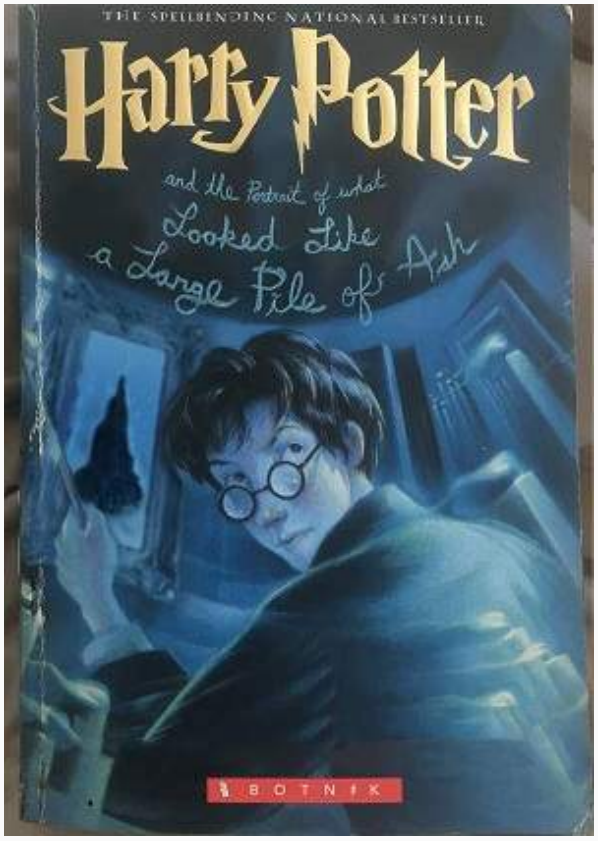
Did you know?

Who do you think wrote Harry Potter and the Portrait of What Looked Like a Large Pile of Ash?

Continue

Harry Potter and the Portrait of What Looked Like a Large Pile of Ash is a Harry Potter novel written by an AI-generated text prediction model (Botnik, no date).

You can learn more about it here: Harry Potter and the Portrait of What Looked Like a Large Pile of Ash by Botnik



Harry Potter and the Portrait of What Looked Like a Large Pile of Ash by Botnik.

Continue

How long do you think it would take to download all the data from the internet?

Continue

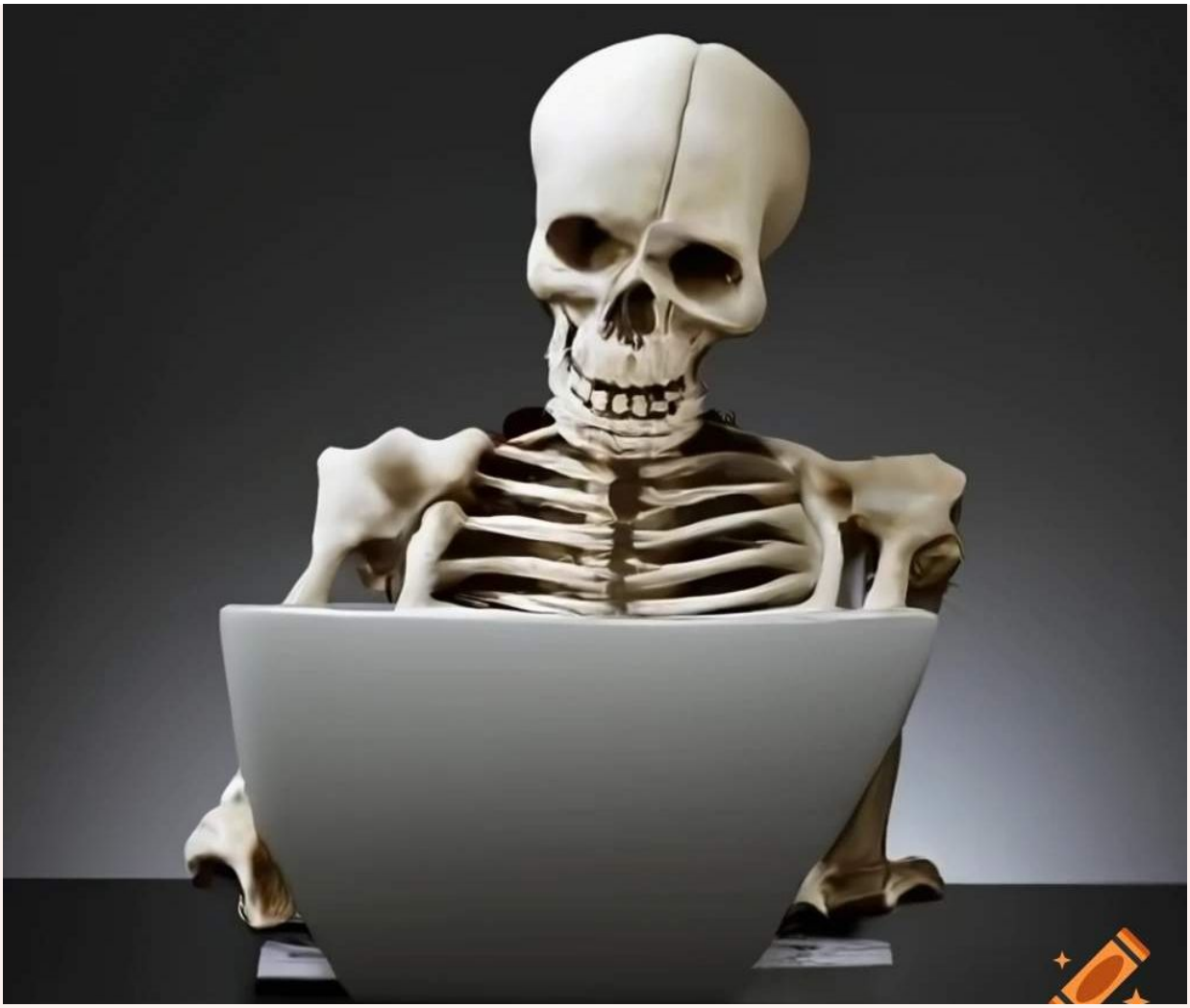


Image created by Craiyon.

It would take 181 million years to download all the data from the internet (Rayaprolu, 2024).

Continue

In this topic, we will conduct a comprehensive review of NumPy and Pandas, both of which you have learned in your previous module, **COM7024 – Programming for Data Analytics**. We will be

looking at some of the fundamental functionalities within these two packages in this topic, but you will continuously be learning these packages as we will use them extensively throughout this module.

Continue

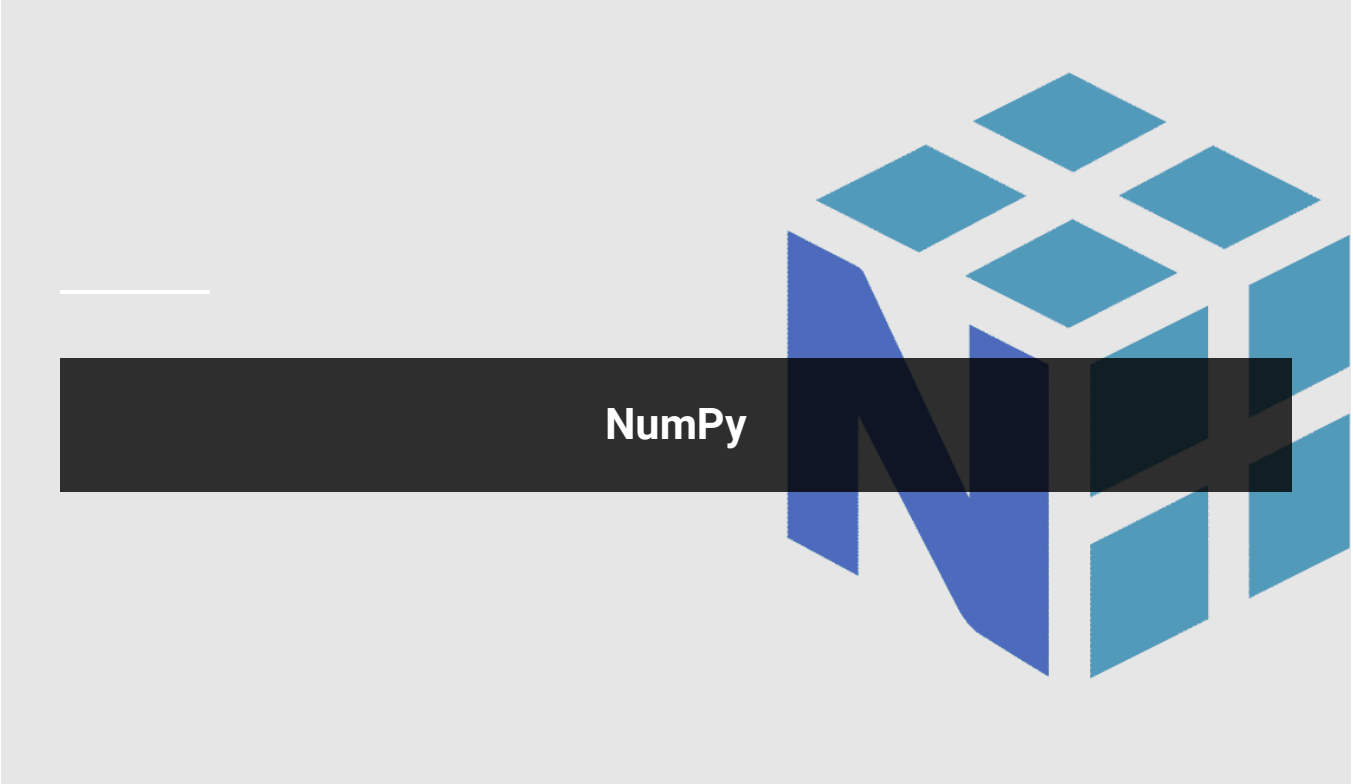


Image from Articulate Rise 360 Content Library.

NumPy (short for Numerical Python) is the fundamental array package in Python. It provides multi-dimensional arrays, high-level functionalities such as linear algebra operations and the Fourier transform, and pseudorandom number generations. Before the introduction of NumPy into Python, some lists served the purpose of arrays, yet they were quite slow. NumPy provides an array object that can be up to 50 times faster.

Scikit-learn Python ML packages use NumPy Array as the fundamental data structure in Python ML packages. Any subset of data that you use as input for your ML model will need to be converted into a NumPy array.

As a revision, let's look at a NumPy array:

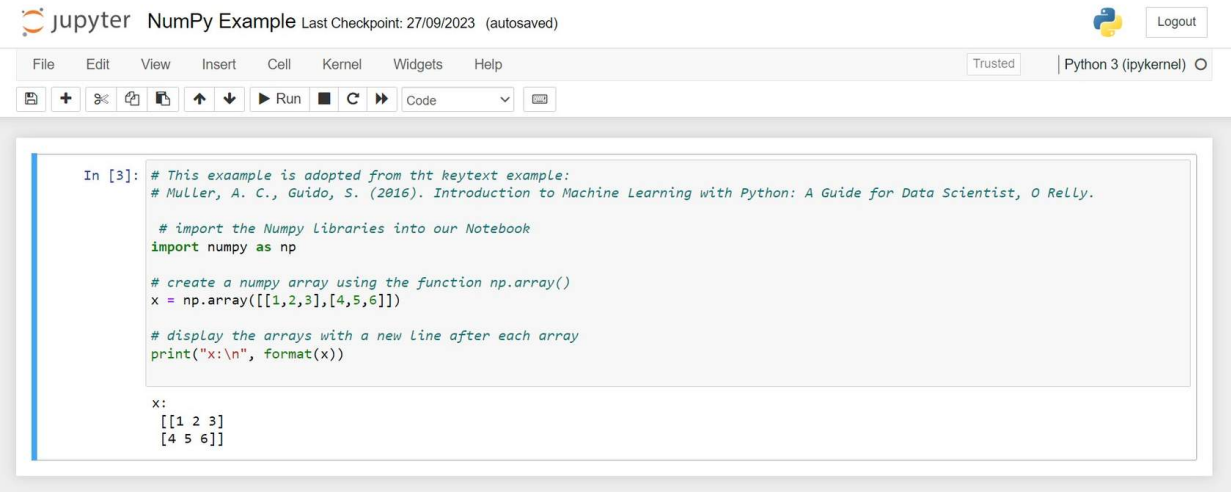


Figure 1.02. NumPy Array in Jupyter Notebook by Arden University. (Accessible code).

Now, we might ask how an array can be used as a data structure for a dataset for ML? Let us examine the output of the program above:

```
# display the arrays with a new line after each array
print("x:\n", format(x))
```

```
x:
 [[1 2 3]
 [4 5 6]]
```

Figure 1.03. NumPy Array output in Jupyter Notebook by Arden University. (Accessible code).

In this example, we have two data instances or rows from a dataset: [1,2,3] and [4,5,6]. They represent the rows and columns we can import into our notebook, which we will look at later in this module. In their native form, they might look like:

Row No	Attribute 1	Attribute 2	Attribute 3
1	1	2	3
2	4	5	6

Figure 1.04. NumPy Array input form by Arden University.

Continue



Image from Pandas.pydata.org.

If we think about NumPy as a blueprint, then Pandas is the library that builds on top of NumPy. Pandas is built around a data structure called DataFrame. In short, DataFrame allows an Excel-line table to be built and provides various functionalities for data wrangling and analysis. Some of the functionalities are:

1. It allows SQL-like queries and join of tables.
2. It allows multi-datatype tables .
3. It allows import of datasets from a variety of file formats.

Let's have a look at an example:

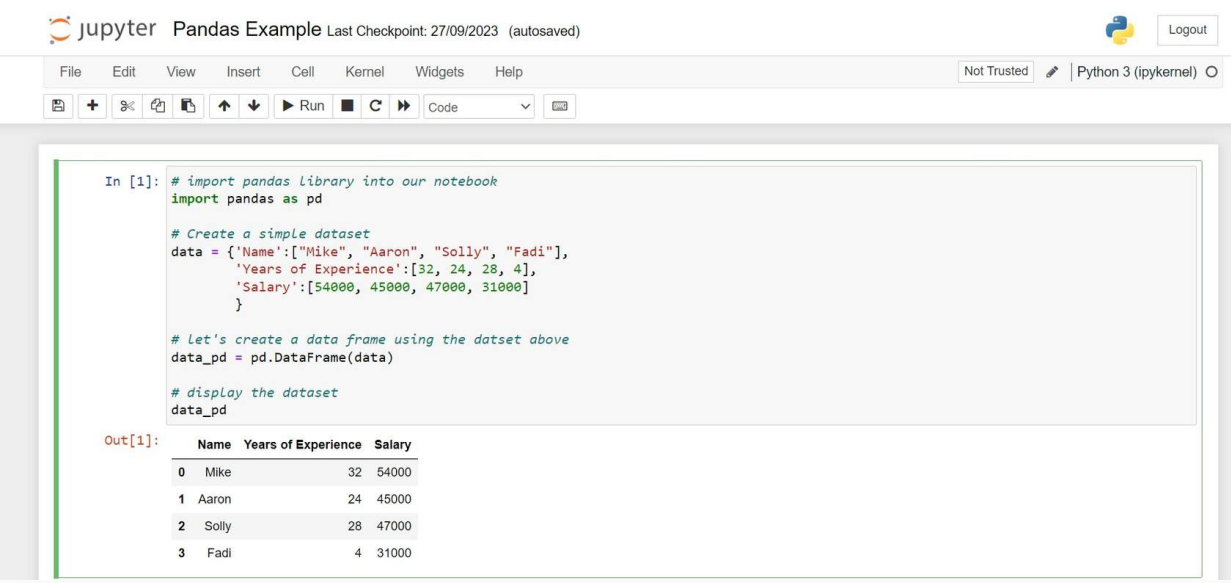


Figure 1.05. Pandas example in Jupyter Notebook by Arden University. (Accessible code).

Next, let's query this table and only display rows that have a salary of more than £45,0000.

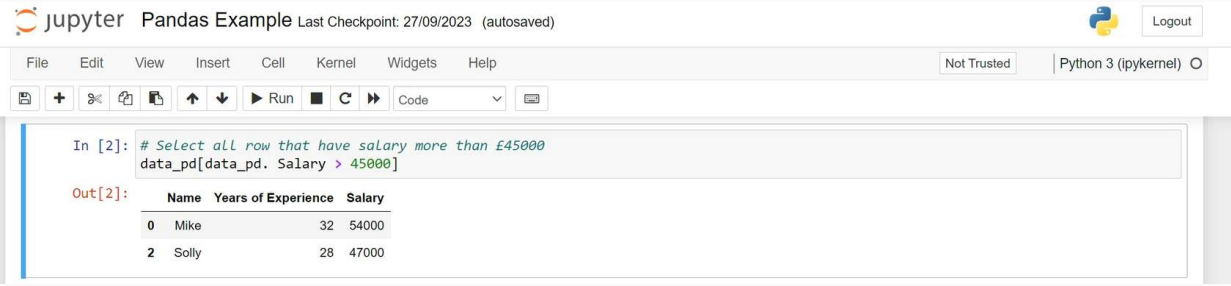


Figure 1.06. Pandas example query in Jupyter Notebook by Arden University. (Accessible code).

Continue

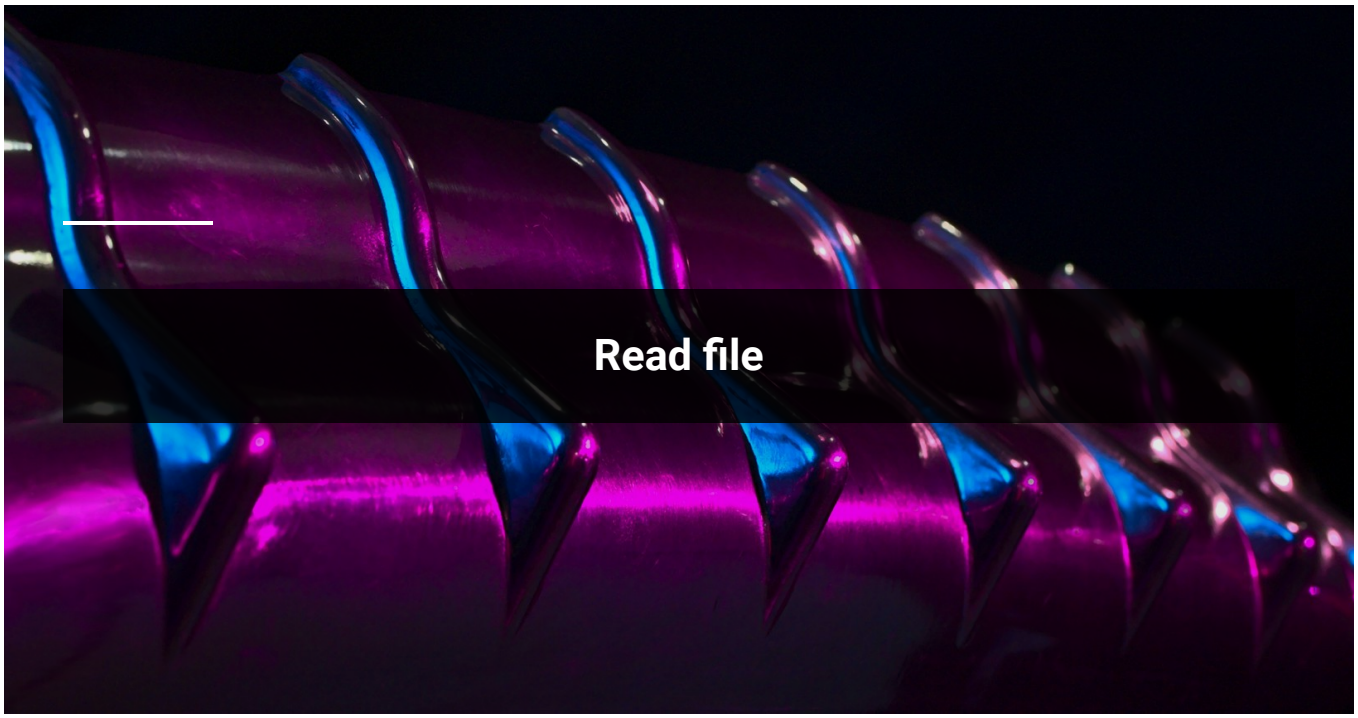


Image from Articulate Rise 360 Content Library.

One of the functions of Pandas that you will be using repeatedly in this module is reading/importing a dataset into your notebook. Pandas can read and write onto various file formats, such as:

- Comma-separated values (CSV) files
- Excel files
- JSON files
- HTML files
- SQL files
- Pickle files

Let's have a look at an example:

Assume we have the data file, in this case Work.csv, uploaded onto the same folder as the notebook you have created.



Figure 1.07. Pandas example dataset file in Jupyter Notebook by Arden University.

We can now load the dataset onto our notebook:

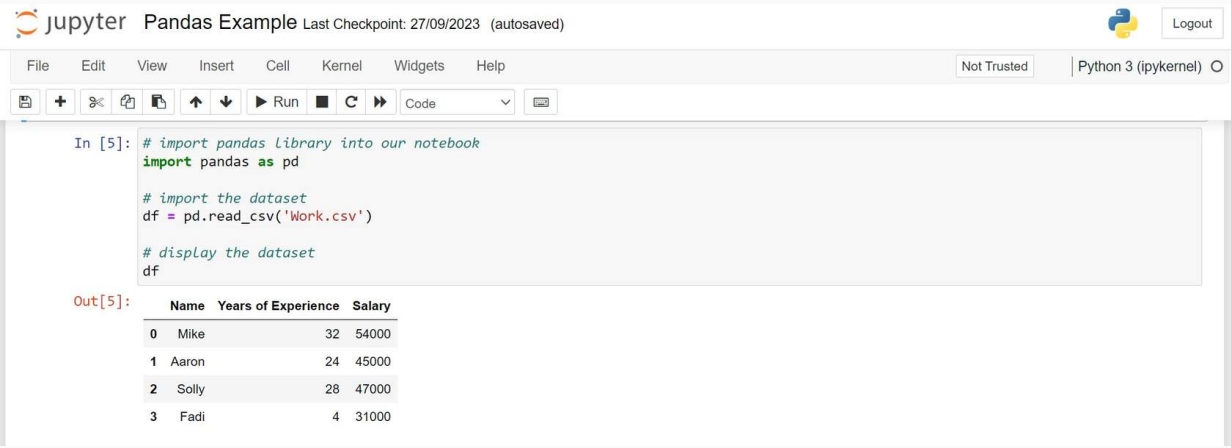


Figure 1.08. Pandas example data read in Jupyter Notebook by Arden University. (Accessible code).

Continue

Topic 3 - Activity 1

Create a cheat sheet (30 minutes)



Create and update a cheat sheet for NumPy and Pandas throughout the module. Use your previous module, **COM7024 – Programming for Data Analytics**, as the starting point.

You can use the below Notepad to help you with this activity, but feel free to use whichever note-taking tool you prefer.

If you decide to use the below Notepad, please remember anything you type in the box below will not be kept when you leave the lesson, so you will need to download your notes for revision and discussion purposes.

My Notes

Input Your Notes Here

Continue

Topic 3 - Activity 2

Reading



The NumPy and Pandas documentation note some essential functionalities you might consider useful in using Python for ML. They are available from:

NumPy

pandas - Python Data Analysis Library

Chapter 1: Big Data, Big Benefits from The Big Data-Driven Business: How to Use Big Data to Win Customers, Beat Competitors, and Boost Profits by Glass and Callahan.

Continue

TOPIC REVIEW

This short topic on NumPy and Pandas has been an opportunity to review what we learned in **COM7024 – Programming for Data Analytics**. In this topic, we looked at the fundamental functionalities of both NumPy and Pandas, and we examined some examples of these functionalities. However, we will continue building our skills and knowledge of these two packages throughout this module.

Continue to Topic 4

Topic 4: Scikit-Learn and SciPy

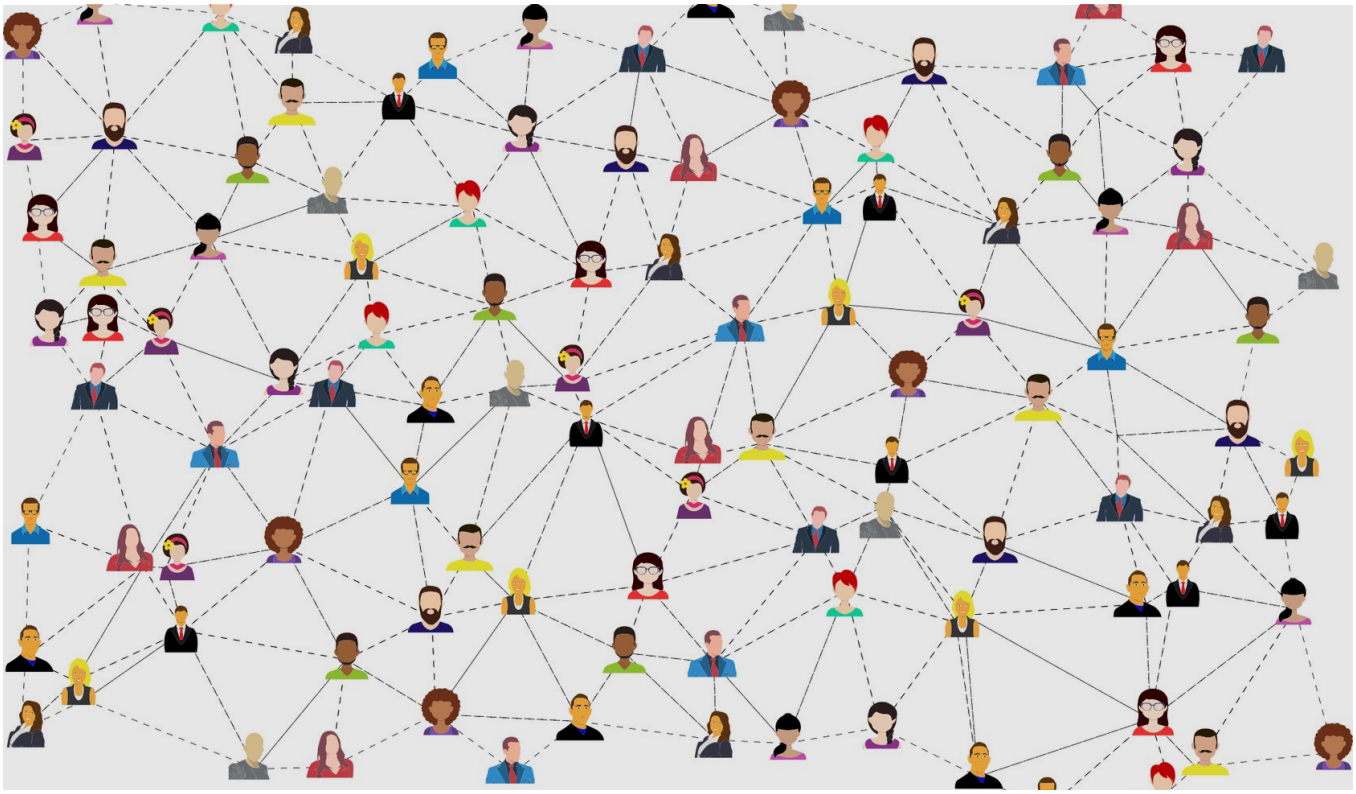


Image from Articulate Rise 360 Content Library.

Did you know?

Scikit-learn and many other packages within Python are meritocratic, consensus-based community projects, and anyone who is interested in contributing to the development is welcome (scikit-learn,

N.D.a).

Continue

ML software shows 93% accuracy in reading lips compared with 52% for human lip-reading experts (Gillespie, 2016).



Image by Sam Loyd. Lip reading puzzle.

Continue

In the final topic of this lesson, we will introduce the main core packages you will use in this module to build ML models using the theory and knowledge you have learned.

As the ancient saying states: 'A craftsman must sharpen his tools to do his job.' So, let us first ensure we have the tools (Scikit-learn and SciPy) in place in Anaconda Jupyter Notebook.

Continue



Image from Github.com.

The default installation of Anaconda Distribution should come with both packages. However, as data scientists, we are focused on detail and preparation.

First, open your *Anaconda Navigator* and select **[Environment]** on the left menu, as shown in the figure below:

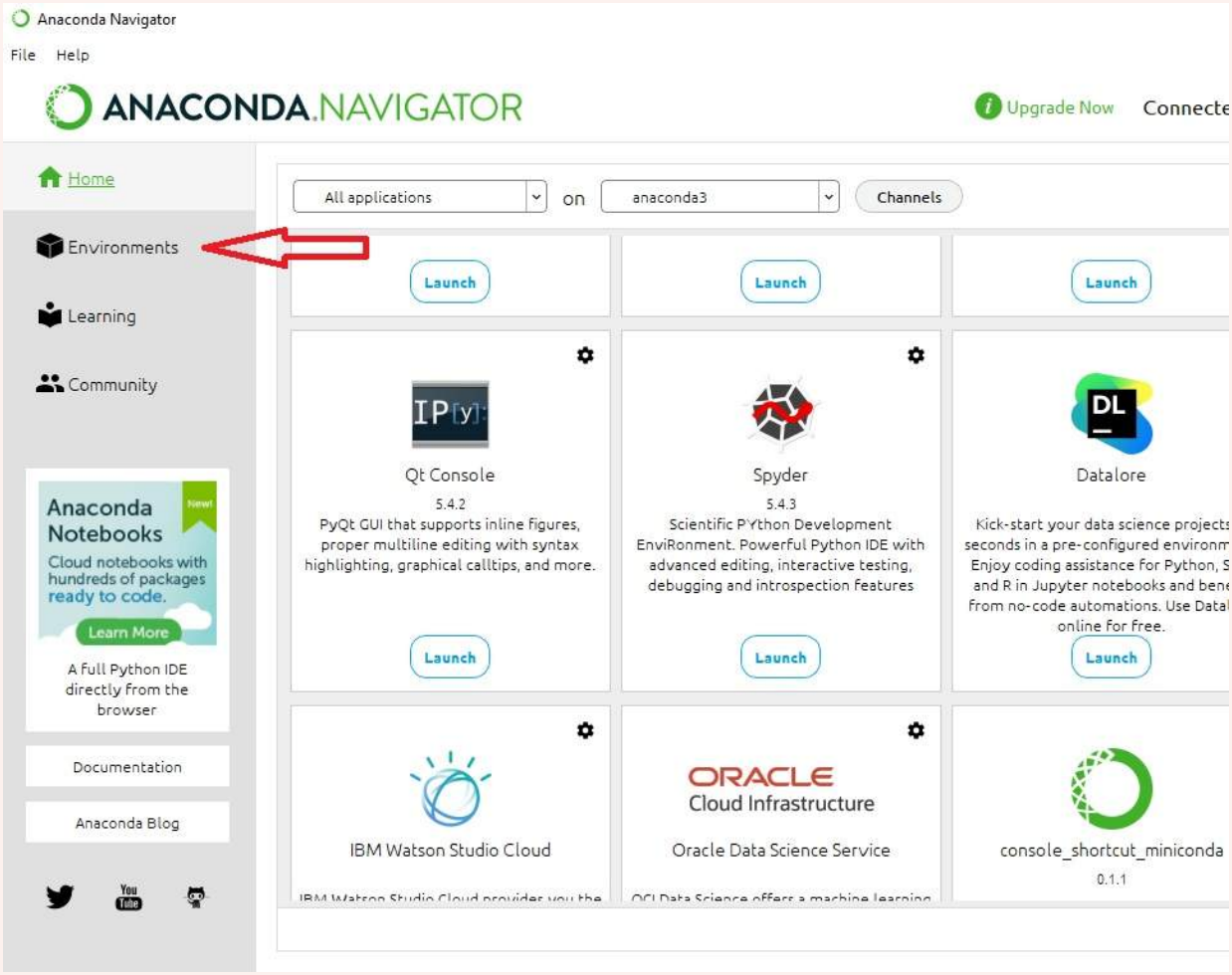


Figure 1.09. Anaconda Navigator environments amended from Anaconda Distribution.

In the Environment screen, please select **[Installed]** from the dropdown box and type in Scikit-learn and Scipy into the **[Search Packages]** box as shown below:

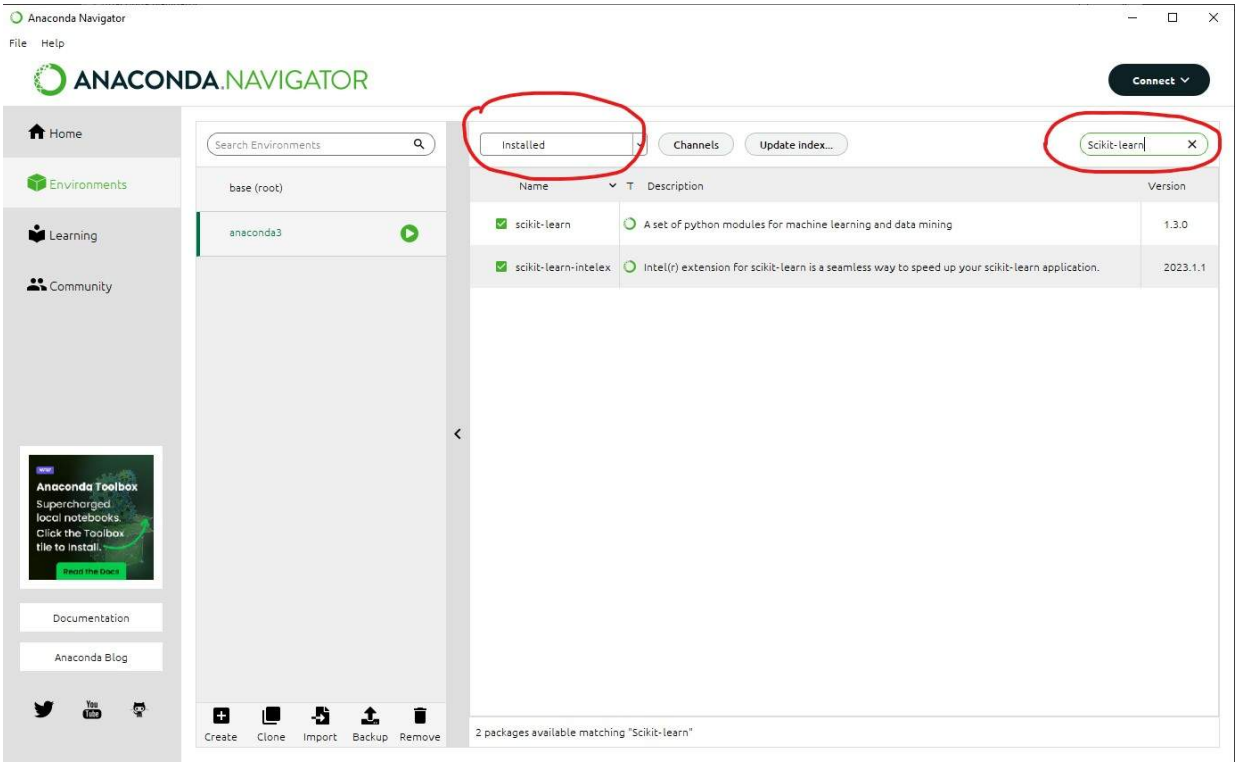


Figure 1.10. Anaconda Navigator package search amended from Anaconda Distribution.

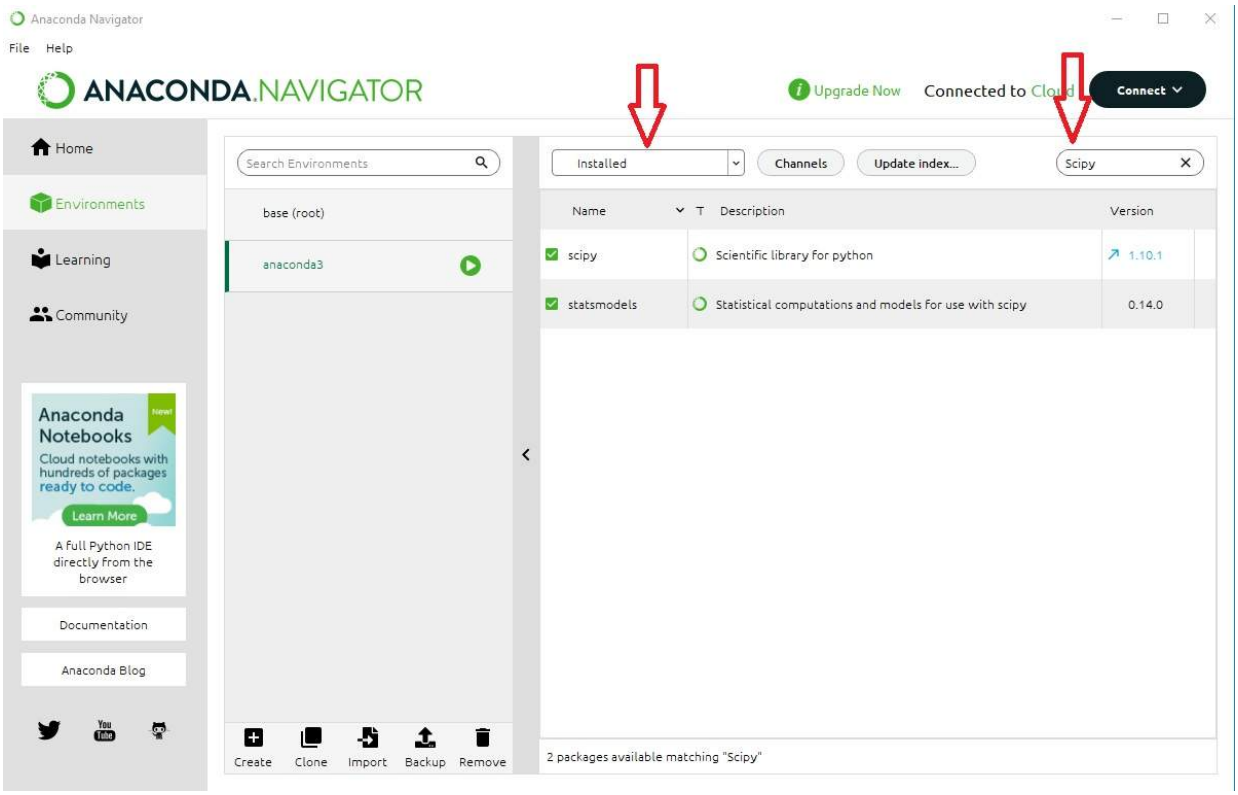


Figure 1.11. Anaconda Navigator package search amended from Anaconda Distribution.

We can see from the figures that both packages are installed.

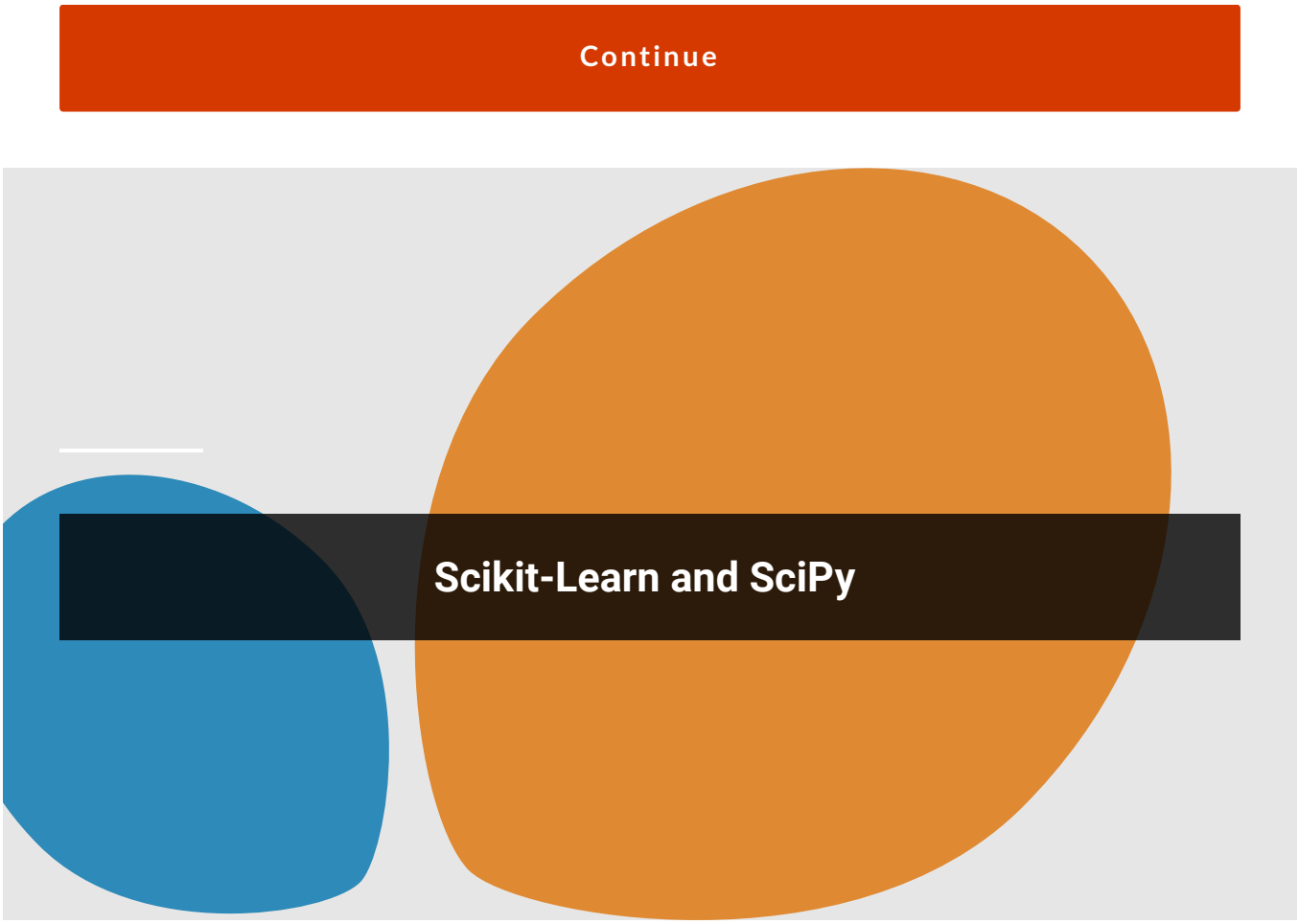


Image from Github.com.

Scikit-learn, together with SciPy, NumPy, Pandas and Matplotlib (Visualisation), forms the full fundamental package for us to learn ML implementation for data analysis.

Scikit-learn is a community-based project that is free and welcomes contribution from all community members. Anyone can access the source code, and it provides the opportunity for further development and improvement. As a tool, it is widely used in industry and academia. It has numerous examples and free online tutorials. It contains all the well-known ML algorithms that we will be learning in this module.

SciPy (short for Scientific Python) consists of a collection of functions for scientific computing. Among the functions, SciPy provides advanced linear algebra functions, signal processing, and mathematical optimisation. Scikit-learn draws its algorithms from SciPy functions.

Continue

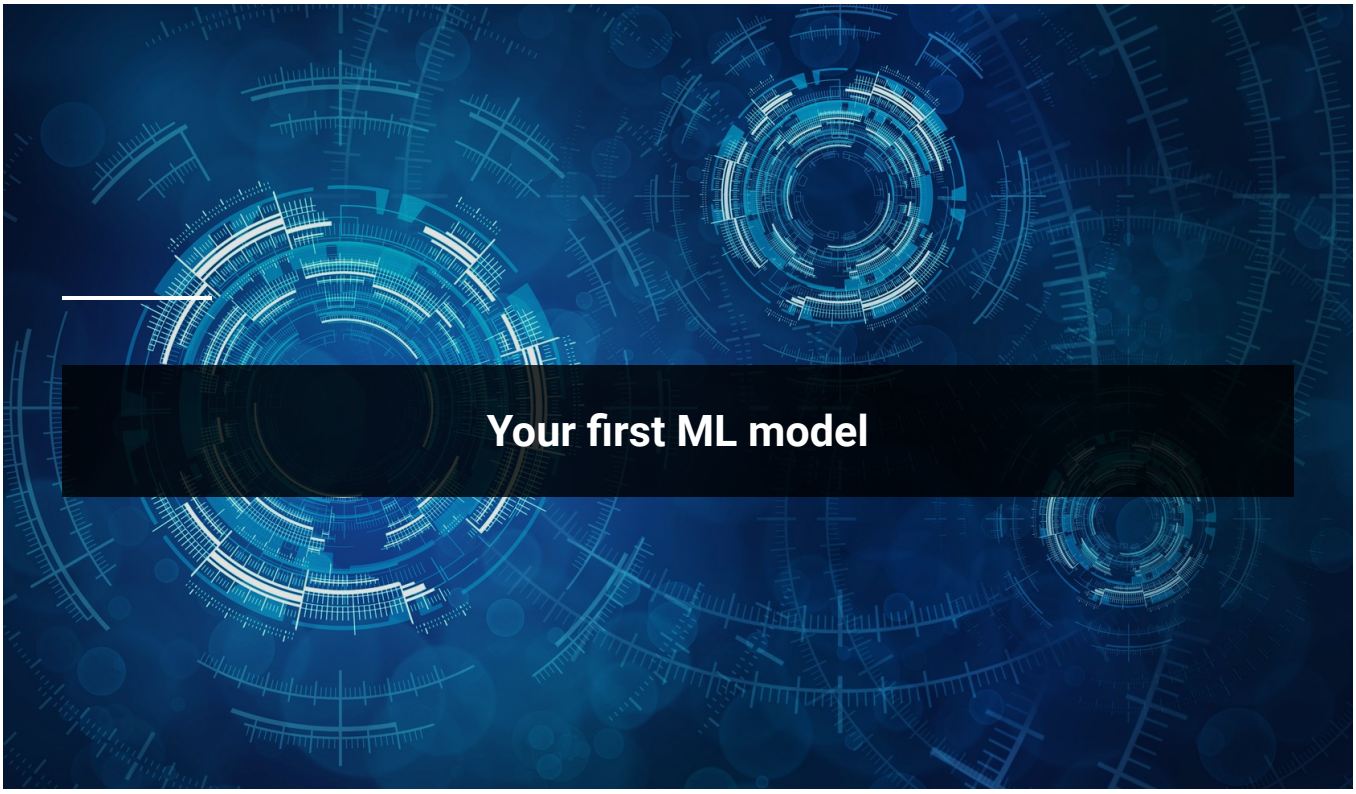


Image from Articulate Rise 360 Content Library.

Because we will cover these two packages extensively in this module, you will not be overloaded with all the functionalities in Lesson 1. Instead, let us look at these packages in action with an example of your first ML model.

Let us create a decision tree. You might not understand all the functions, but the purpose of this example is to provide you with a first taste of how an ML model is implemented.

We will use the Iris dataset from Fisher, which is a very popular dataset for data analysis. The full description of the dataset can be found here: Iris by UCI Machine Learning Repository

Please go through the description, which will give you a better understanding of the data.

The ML model called the decision tree is deployed in this example to decide if an unknown Iris data belongs to one of three types of varieties: ***Iris Sentosa***, ***Iris Versicolor*** or ***Iris Virginica***.

The following are the steps we will follow. When we dive into the module further, the steps do not change much.

1. Import necessary libraries.
2. Load and pre-process the dataset.
3. Split the data into training and testing sets.
4. Create a decision tree classifier and train it on the training data.
5. Evaluate the model on the testing data.
6. Visualise the decision tree.

The program code is as follows:

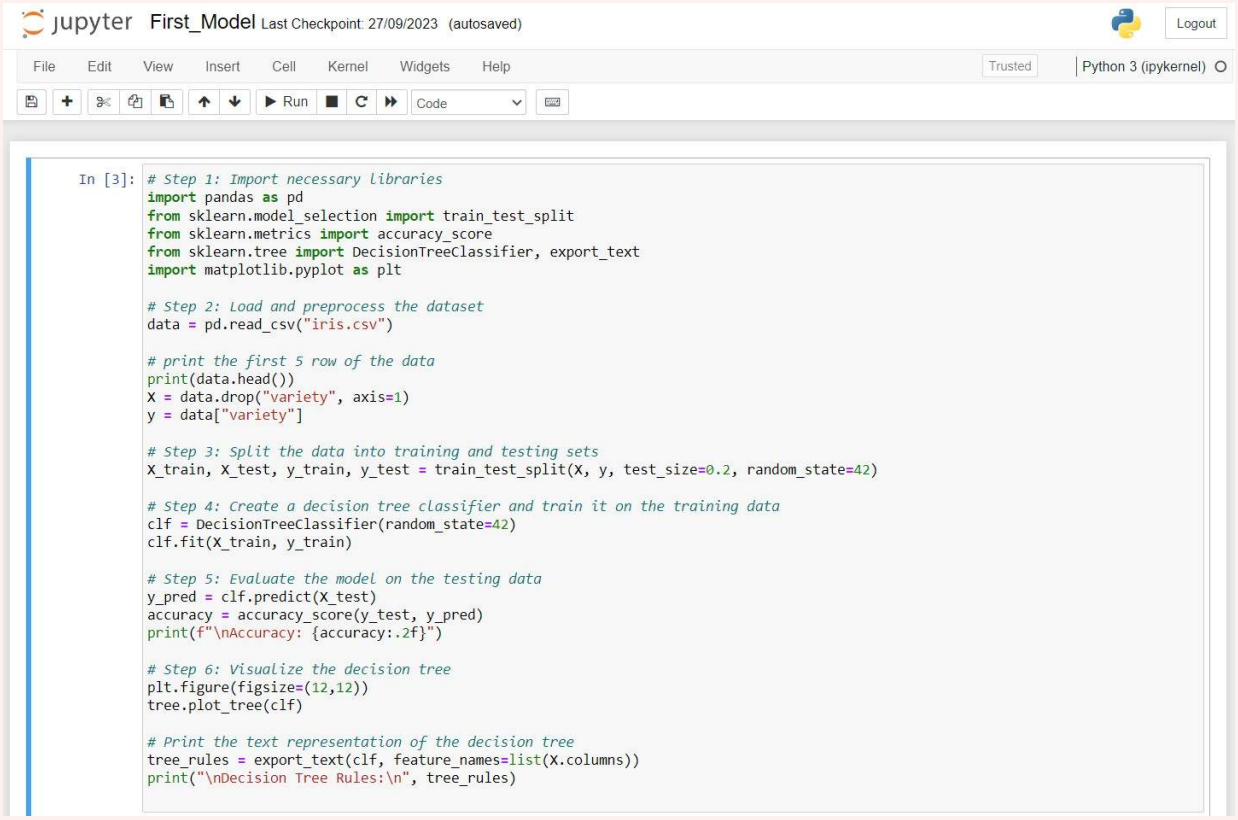


Figure 1.12. First machine learning model in Jupyter Notebook by Arden University. (Accessible code).

The output of the program:

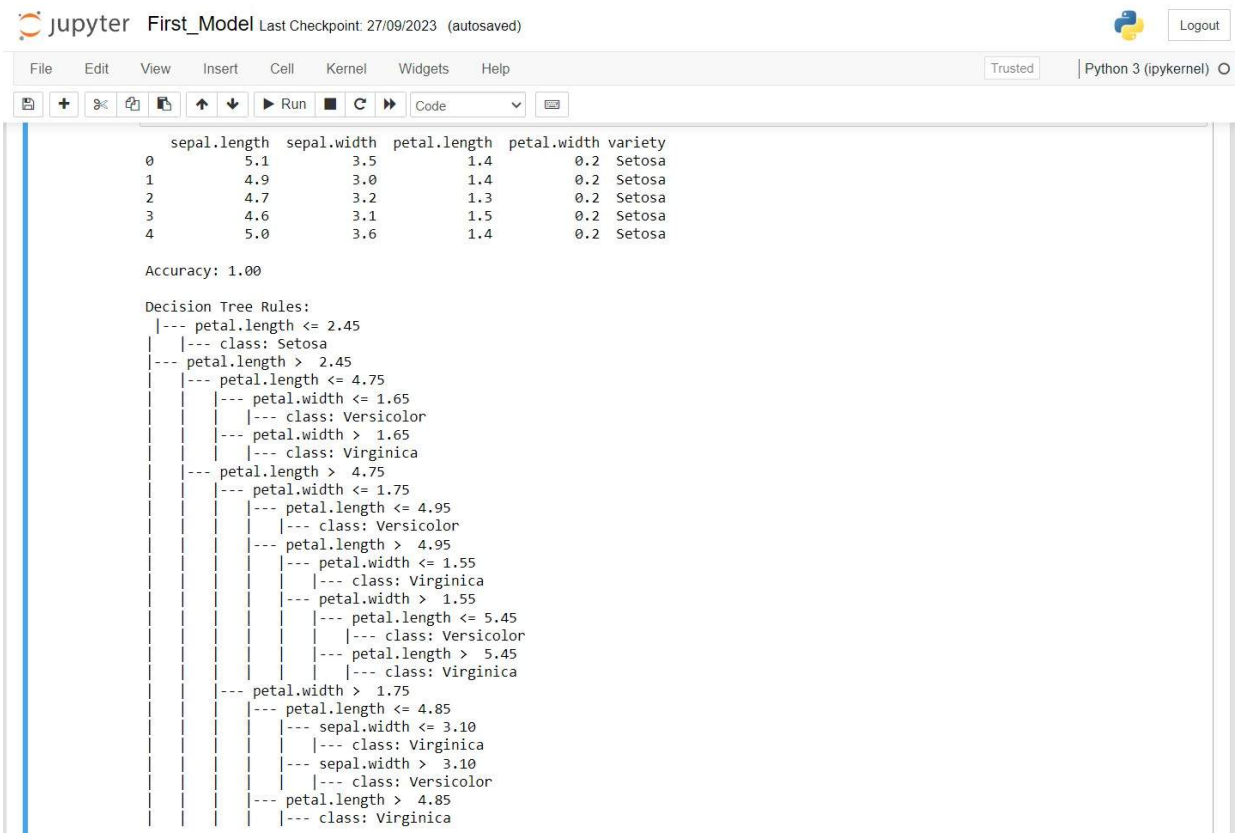


Figure 1.13. Machine learning model output in Jupyter Notebook by Arden University. (Accessible text).

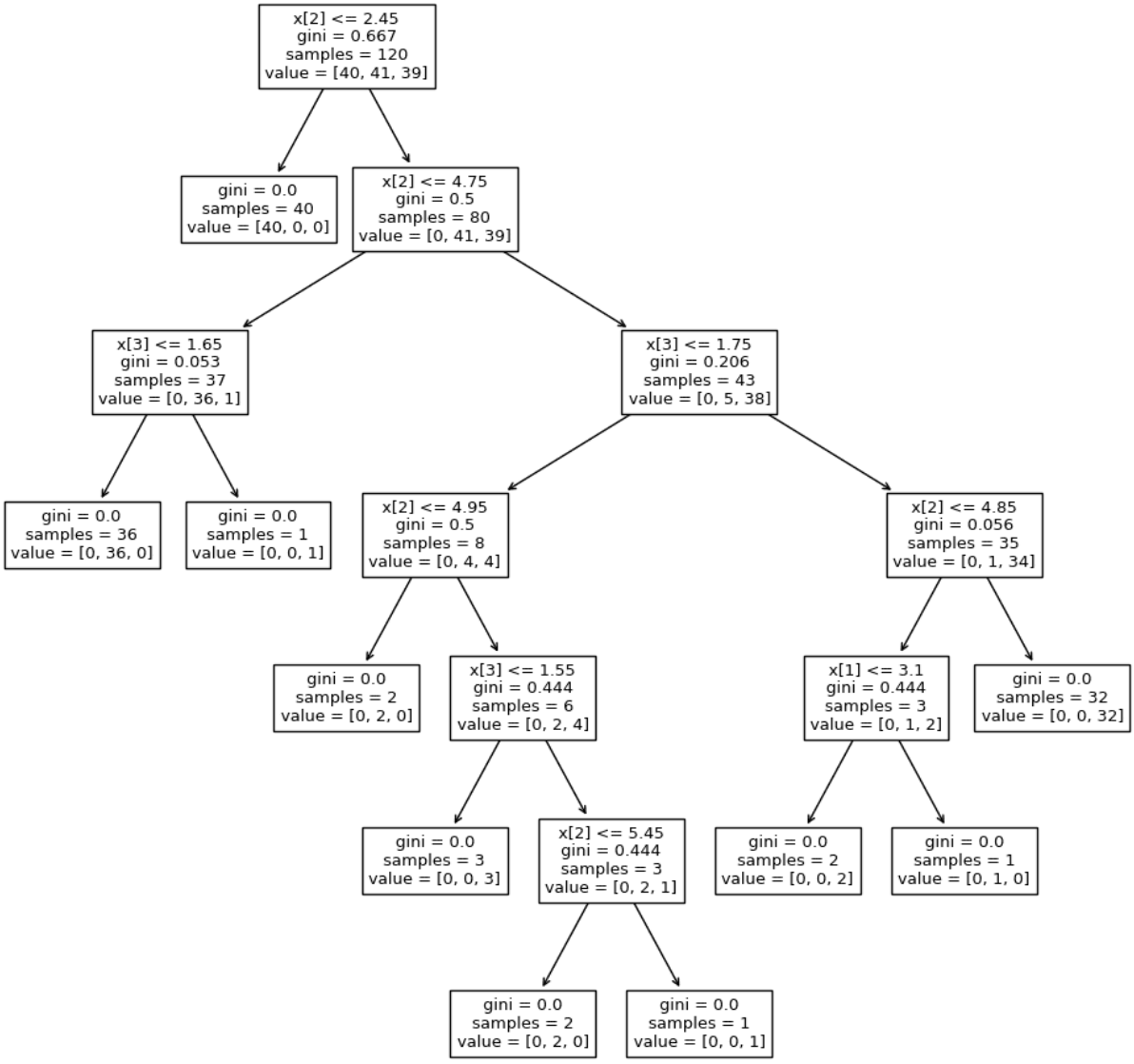


Figure 1.14. Machine learning model output – Decision Tree in Jupyter Notebook by Arden University.

We will dive deeper into decision trees in Lesson 5 of the module.

Continue

Topic 4 - Activity 1

Discussion on iLearn - Understanding the functions of Scikit-learn and SciPy (30 minutes)



What are the main functions of Scikit-learn and SciPy? Please post your answers to the group discussion forum.

Continue

Topic 4 - Activity 2

The Scikit-learn and SciPy documentation includes some essential functionalities when using Python for ML. Please visit the websites below and state within the group discussion forum how these relate to what you have learned within the topic so far.

- scikit-learn
- SciPy



Continue

Topic 4 - Activity 3

Reading



Chapter 1: Introduction from Introduction to Machine Learning with Python: A Guide for Data Scientists by Müller and Guido.

Continue

TOPIC 4: SUMMARY

This final topic provided an introduction to Scikit-learn and SciPy, two packages that you will use extensively in your journey to becoming an ML practitioner. As a taster, you saw all four of the packages we introduced in this lesson in action by building a decision tree to classify the variety of Iris data.

Continue to Knowledge check

Knowledge check

Select the correct answer for the following questions.

1 of 3: We live in an age of data explosion.
Which of the following is **TRUE**?

- When data increases, knowledge increases.
- When data increases, knowledge decreases.
- When data decreases, knowledge increases.
- When data decreases, knowledge decreases.

SUBMIT

Continue

2 of 3: Which of the following are the machine learning types?
Select all of the correct answers.

- [ ] Supervised learning
- [ ] Statistical learning
- [ ] Unsupervised learning
- [ ] Reinforcement learning

SUBMIT

Continue

3 of 3: Which of the following are applications of machine learning?
Select all of the correct answers.

- [ ] Predict the sales for next year
- [ ] Sorting letters using postcodes
- [ ] Retrieve the sales figure from last year
- [ ] Product recommendation on Amazon

SUBMIT

Continue to Lesson summary

Lesson summary

LESSON SUMMARY

Well done, you have completed Lesson 1!

Lesson 1 has been an introduction to the world of machine learning for data scientists, discussing the meaning of ML and the challenges it poses to an ML practitioner. In the first topic, we introduced the concept of ML and discussed the type of learning and real-world applications of ML.

Topic 2 explored the essential toolbox that every budding ML analyst should have. Some of these individual tools you may already be confident with, and some not. Either way, recapping and solidifying fundamental knowledge are essential. Without the foundations to any endeavour or process, one cannot progress to the more complex processes available, especially within a data analysis environment.

In Topic 3, we provided a general review of two packages, NumPy and Pandas. These are the fundamental libraries in Python when dealing with data, which is an essential part of becoming an ML practitioner and data scientist in general. It also links what we have learned in our previous module, **COM7024 – Programming for Data Analytics**, and how we can contextually build our skills onto machine learning environments.

Finally, in Topic 4 we discussed the main packages, Scikit-learn and SciPy, that we will use in this module for developing our hands-on practical skills and for building

on the ML algorithms we will be learning. We concluded this lesson by looking at a simple ML model as a taste of what is to come.
Having completed all the topics in this introductory lesson, hopefully you have some insight into the many benefits and challenges you might face when studying this module. With the background knowledge being completed, we can now progress to Lesson 2 where we will look at the input of all ML models: data!

Continue to 'Reading list'

Reading List

All reading and resources in this module are provided by the Arden University Library service. The full reading list for this module can be accessed at the link below. More books, journals and databases can be found on the Library Portal. If you need these resources in a different format, our accessibility tools can help.

Full reading list

You can find all the key, essential and wider reading and resources for this module on the dedicated reading list page on the Learning Success Hub.

VIEW READING LIST

References

Cited sources that do not appear in your reading list:

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